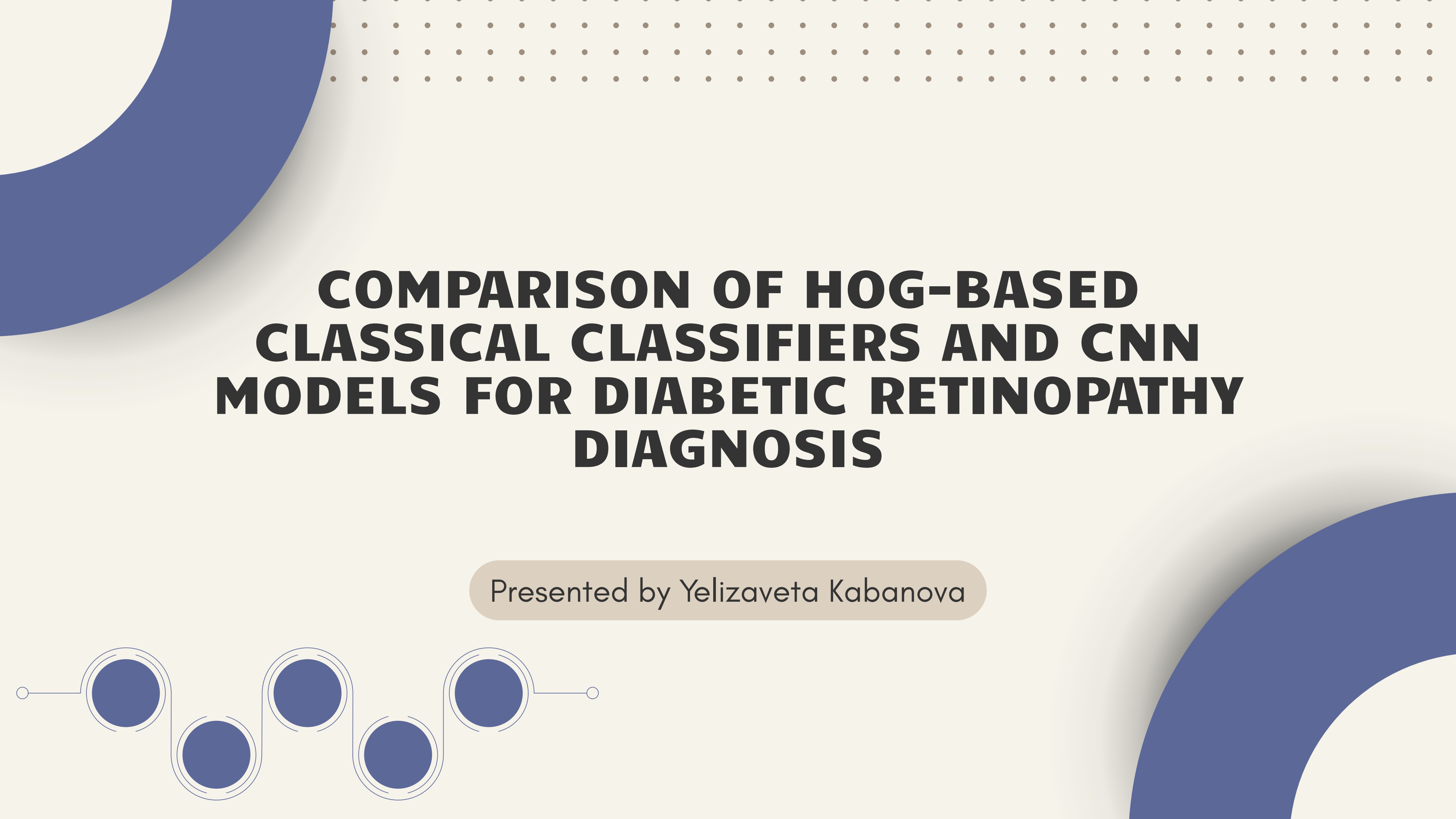
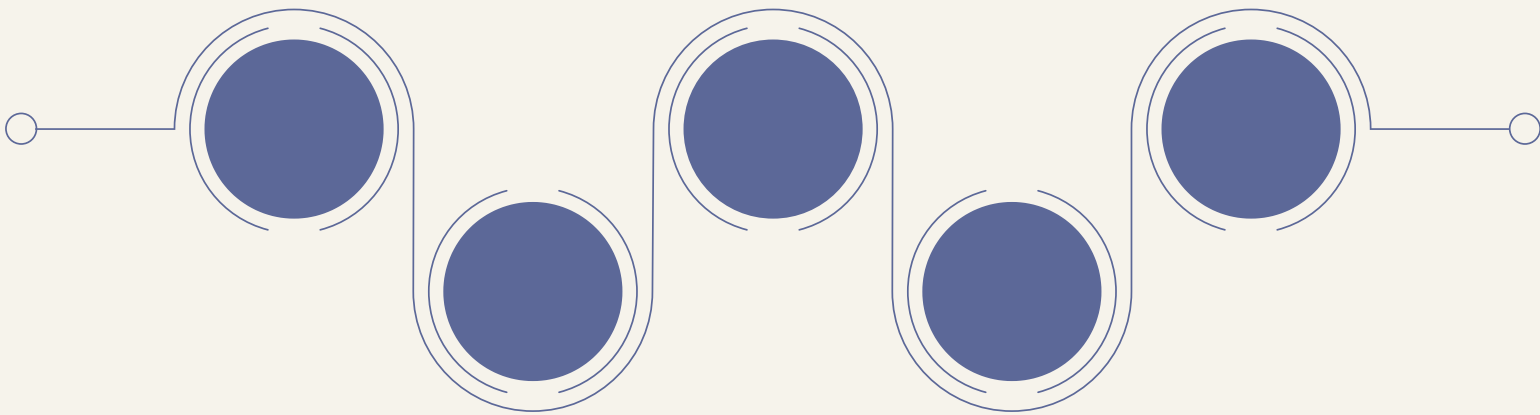




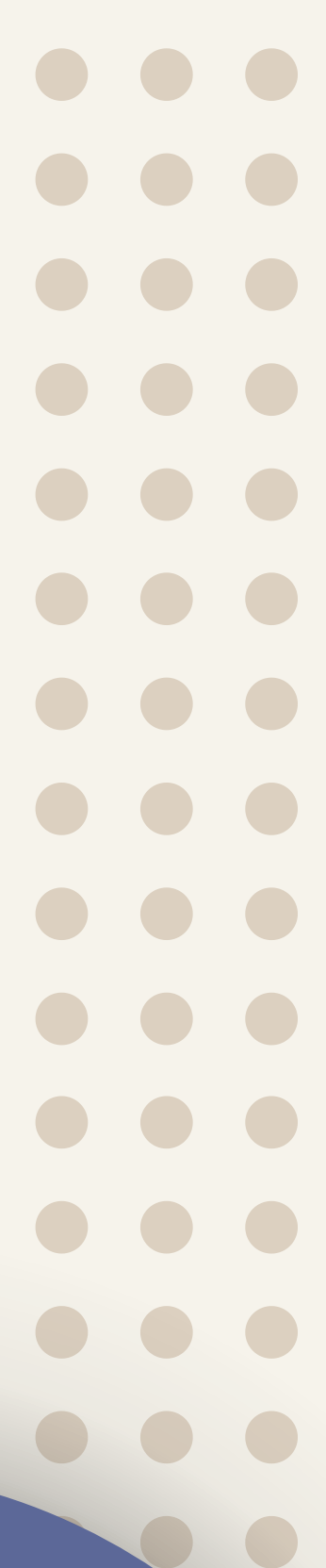
COMPARISON OF HOG-BASED CLASSICAL CLASSIFIERS AND CNN MODELS FOR DIABETIC RETINOPATHY DIAGNOSIS

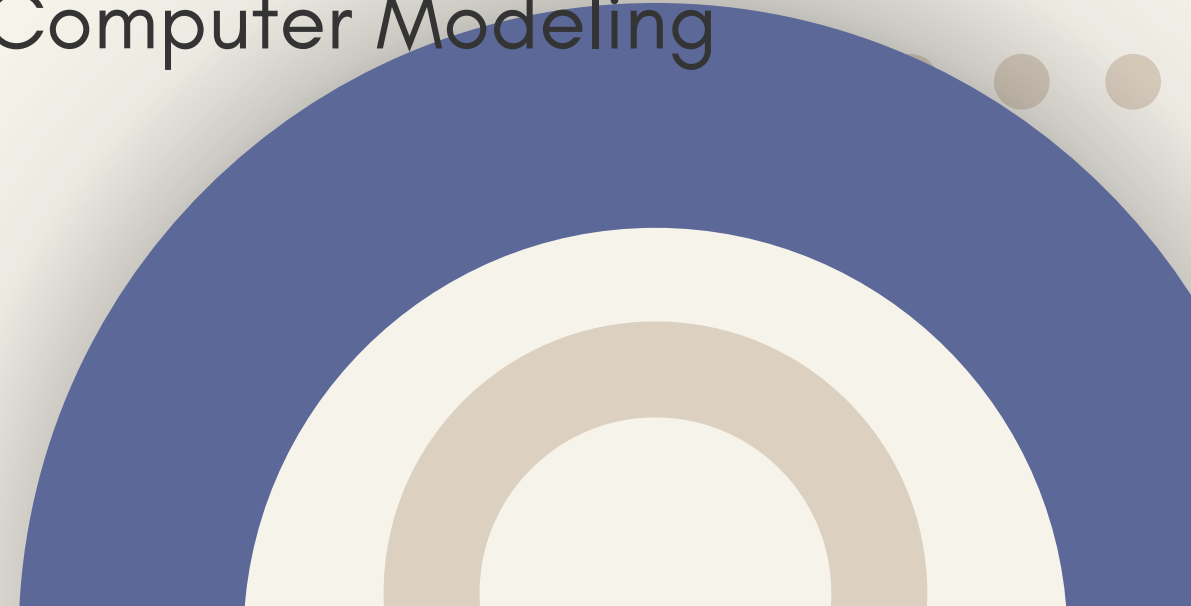
Presented by Yelizaveta Kabanova





Introduction



- **Title:** Comparison of HOG-based Classical Classifiers and CNN Models for Diabetic Retinopathy Diagnosis.
 - **Authors:** Yelizaveta Kabanova, Nataliia Kuznietsova, Kateryna Ivanko, Vishwesh Kulkarni.
 - **Affiliations:** National Technical University of Ukraine "Igor Sikorsky Kyiv Polytechnic Institute"; King's College London; SUSTech-King's School of Medicine.
 - **Conference:** The Ninth International Workshop on Computer Modeling and Intelligent Systems (CMIS-2026).
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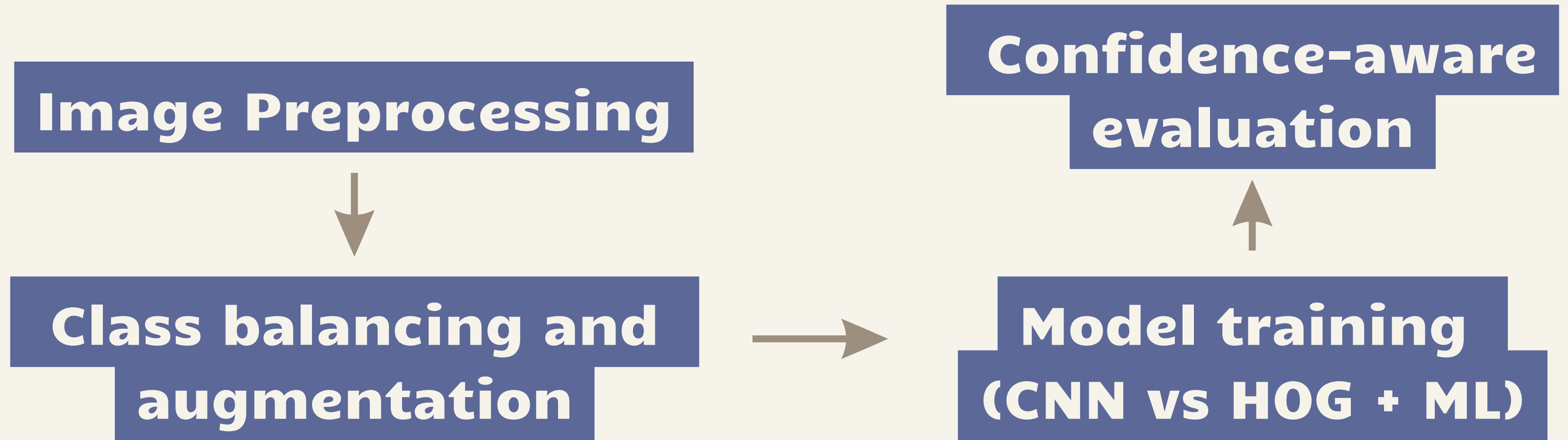
Motivation

- **The problem:** Diabetic Retinopathy (DR) is a major cause of vision loss and requires accurate staging for timely treatment.
- **Medical Necessity:** In medical image analysis, the reliability of model predictions is as critical as accuracy, as incorrect but highly confident predictions lead to serious clinical consequences.
- **The Gap:** Existing studies often neglect predictive confidence and rarely compare classical machine learning with modern CNN architectures under identical conditions.

Objective and Contributions

- **Main Goal:** The study performs a comprehensive, confidence-aware comparison between HOG-based classical classifiers and a CNN architecture.
- **Contribution 1:** A unified evaluation framework for fair comparison under identical preprocessing and balancing conditions.
- **Contribution 2:** Extension of uncertainty analysis to confidence-correctness alignment metrics such as AURC and E-AURC.
- **Contribution 3:** Reliability-centered interpretation of model behavior in DR classification.

Methodology



All models were trained and evaluated under identical preprocessing, class balancing, and evaluation protocols to ensure a fair comparison.

Dataset and Main Challenges

- **Dataset:** APTOS 2019 Blindness Detection (3,662 img, 5 classes).
- **Key Challenge:** Severe class imbalance: Class 0 (No DR) dominates at 49.3%, while Class 3 (Severe) is only 5.3%.
- **Image Issues:** Uneven lighting, black borders, artifacts, and glare.
- **Proposed Solution:** Oversampling + custom augmentations.
- **Train/Validation split:** 80/20 ratio.



Data Preprocessing

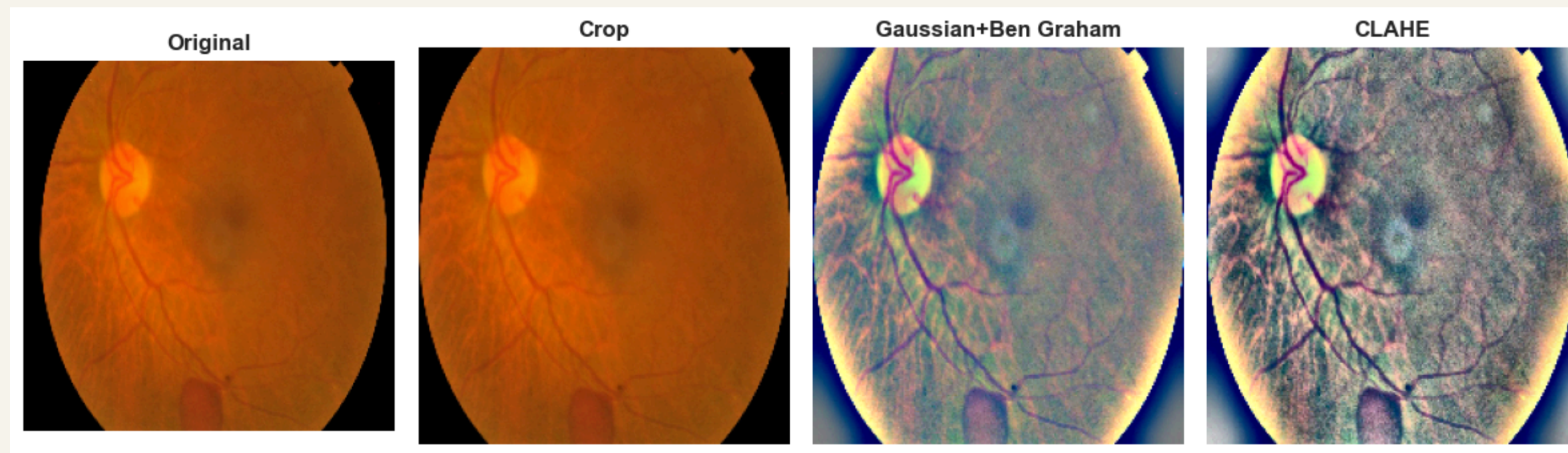
Preprocessing:

- Cropping (removal of black borders);
- Resizing (224×224);
- Normalization ([0,1]);
- Noise reduction (Gaussian blur);
- Contrast enhancement (CLAHE + linear enhancement).

Data Augmentation:

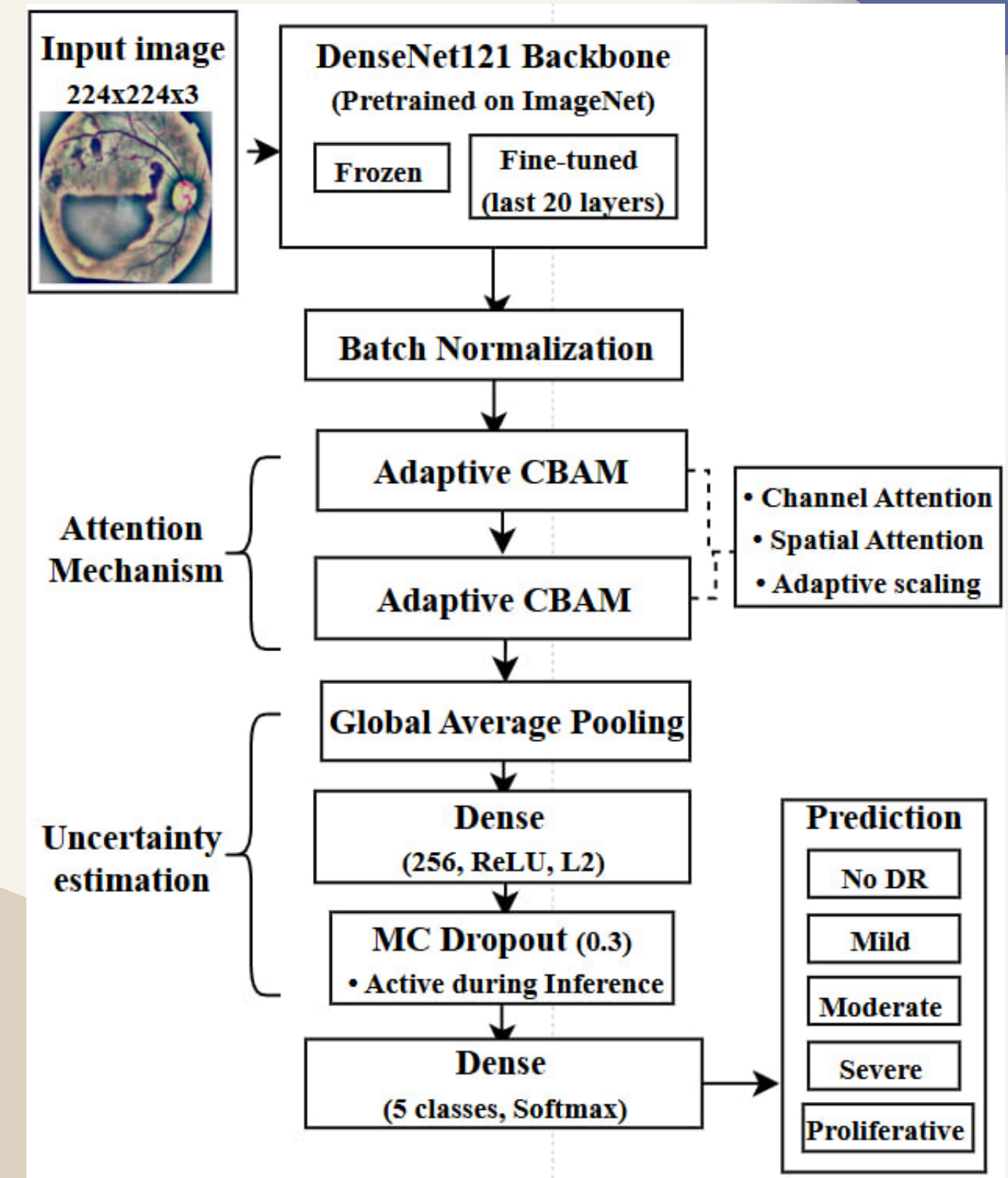
- Balancing stage: flips, brightness ($\pm 20\%$), rotation ($\pm 15^\circ$);
- Training stage: zoom, shifts, flips, brightness, rotation.

The training set was expanded to 9,025 images.



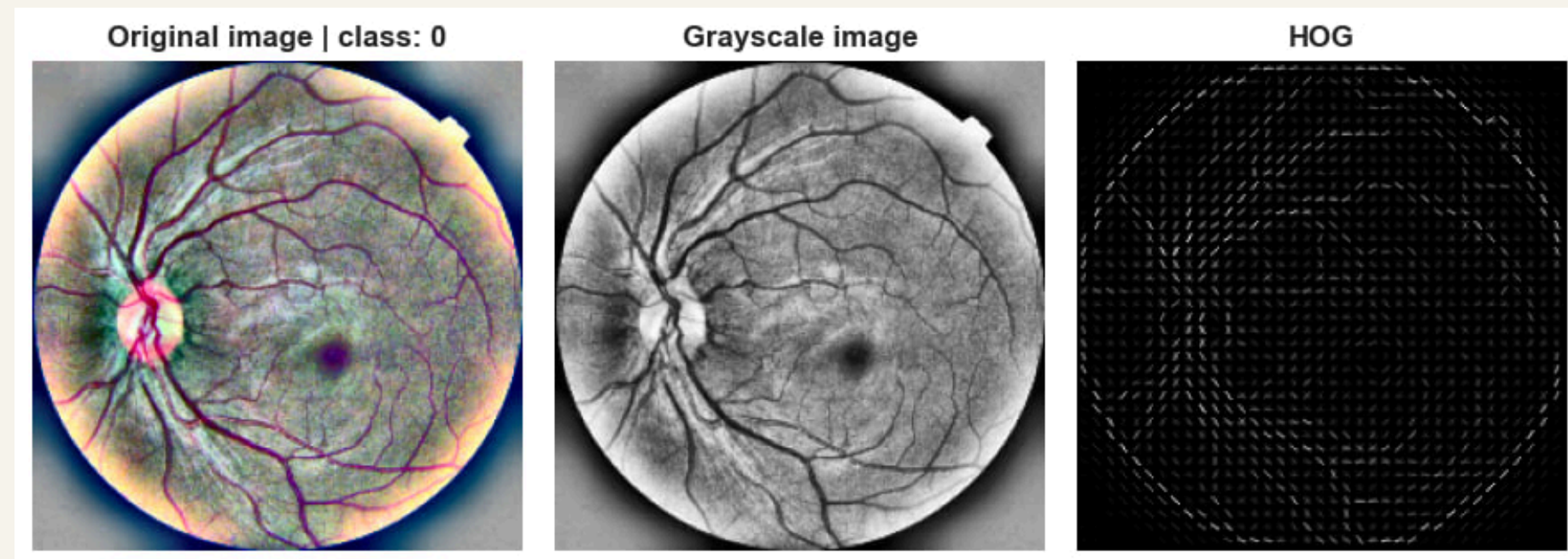
CNN Architecture

- **Backbone:** DenseNet121 (pretrained on ImageNet, fine-tuned last 20 layers).
- **Attention:** Adaptive Convolutional Block Attention Modules (CBAM) based on feature variance.
- **Uncertainty:** Monte Carlo Dropout (rate=0.3) was incorporated during inference for uncertainty estimation.
- **Optimization:** Focal Loss was used to enhance sensitivity to complex, underrepresented examples.
- **Training details:** Adam optimizer, batch size 32, 30 epochs.
- **Learning rates:** 0.0001 (head), 0.00001 (fine-tuning).



Handcrafted Feature Extraction (HOG)

- **The Approach:** Histogram of Oriented Gradients (HOG) encodes local structural information by analyzing gradient distributions within spatial cells.
- **Optimal Config:** 9 orientation bins, 6x6 pixels per cell, 2x2 cells per block.
- **Normalization:** L2-Hys method for robustness.
- **Benefits:** Interpretability and computational efficiency.



Classical ML Classifiers

- **SVM:** Radial Basis Function (RBF) was selected for complex high-dimensional feature spaces, using a One-vs-One (OvO) strategy for multi-class grading.
- **Random Forest (RF):** An ensemble method aggregating multiple decision trees to reduce variance and improve generalization; stability testing (200, 300, 500 trees).
- **Dimensionality Reduction:** Principal Component Analysis (PCA) with 160 components was applied for noise suppression and efficiency.



Evaluation Metrics

- **Standard:** Quadratic Weighted Kappa (QWK), Accuracy, Precision, Recall, F1-score, AUC.
- **Reliability:** AURC (Area Under Risk-Coverage Curve).
- **Error Detection:** Error AURC (E-AURC).
- **Separation:** Difference in confidence between correct and wrong passes.
- **Confidence:**
 - CNN → mean predictive probability via Monte Carlo Dropout;
 - SVM → probability from auxiliary One-vs-Rest classifier;
 - RF → maximum predicted class probability.

QWK was selected as the main metric because it accounts for the ordinal nature of diabetic retinopathy stages.



Results



CNN:

- Achieved the highest accuracy (**QWK=0.92, F1=0.85, AUC=0.97**) and most reliable uncertainty estimates (AURC=0.03).
- Demonstrated more stable performance across all classes, including minority classes.

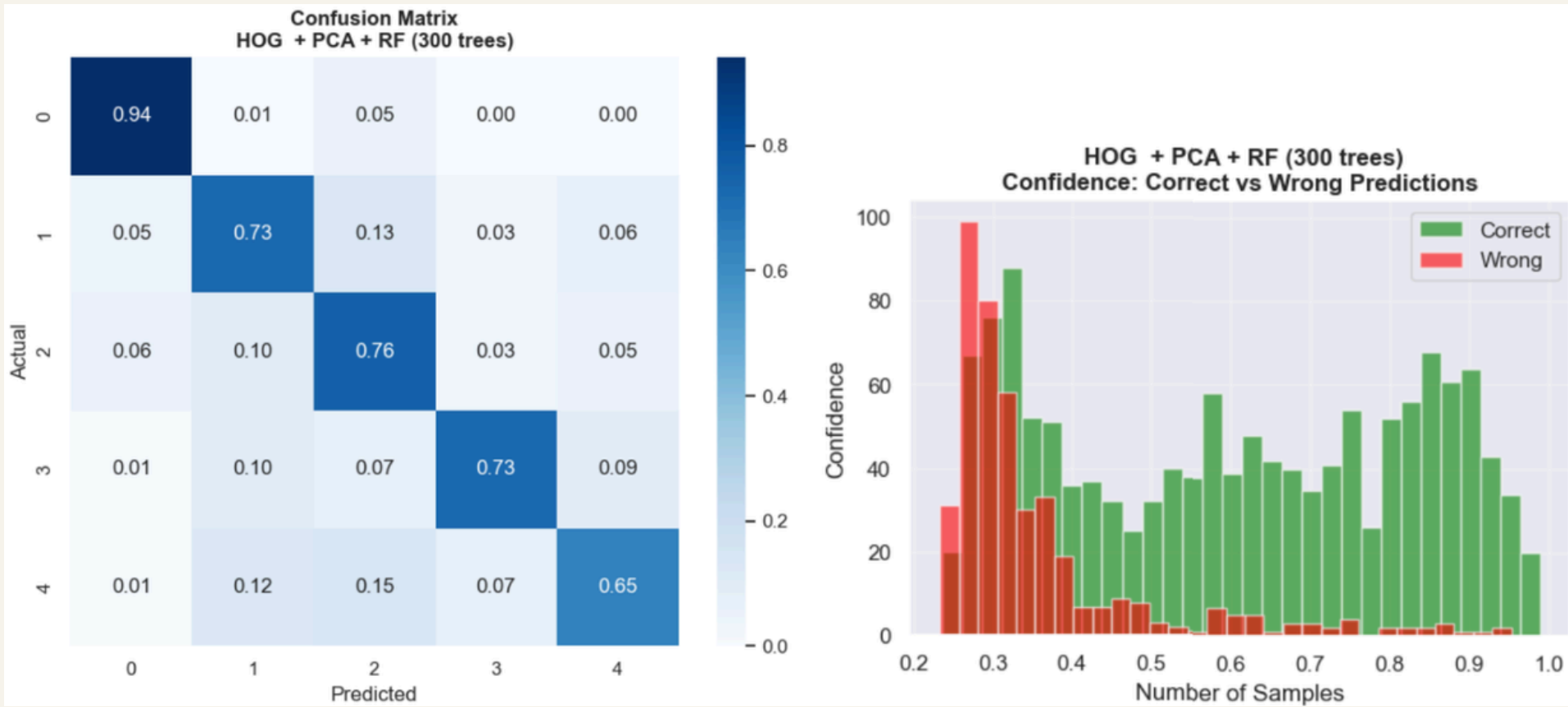
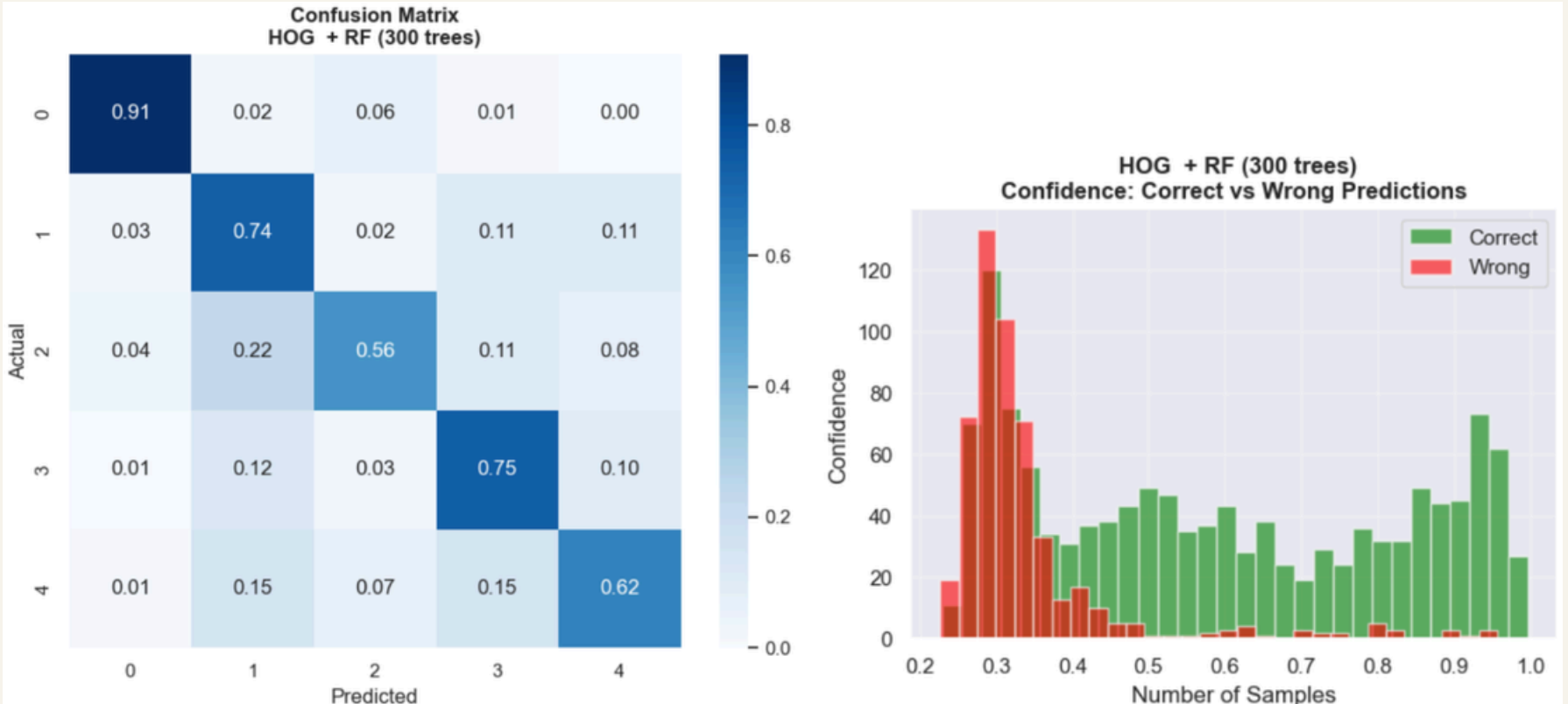
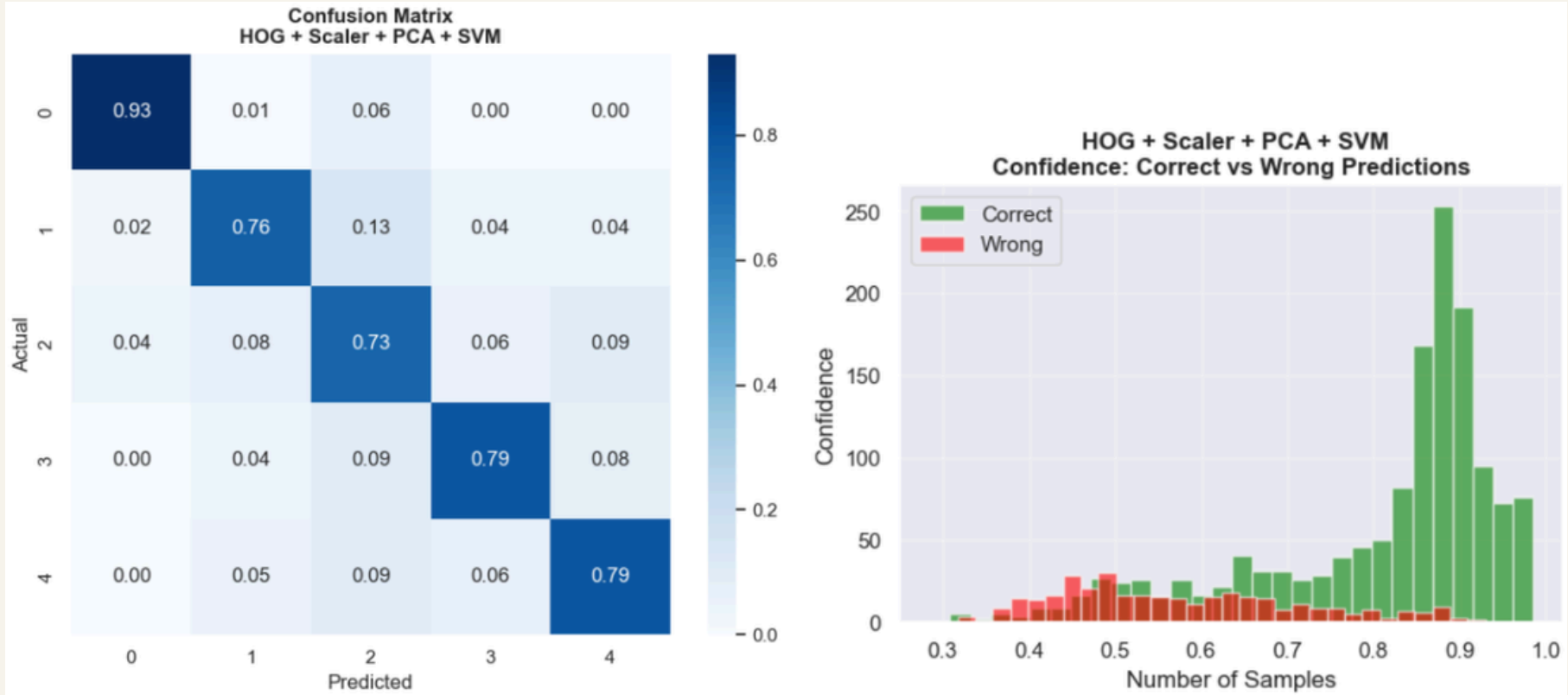


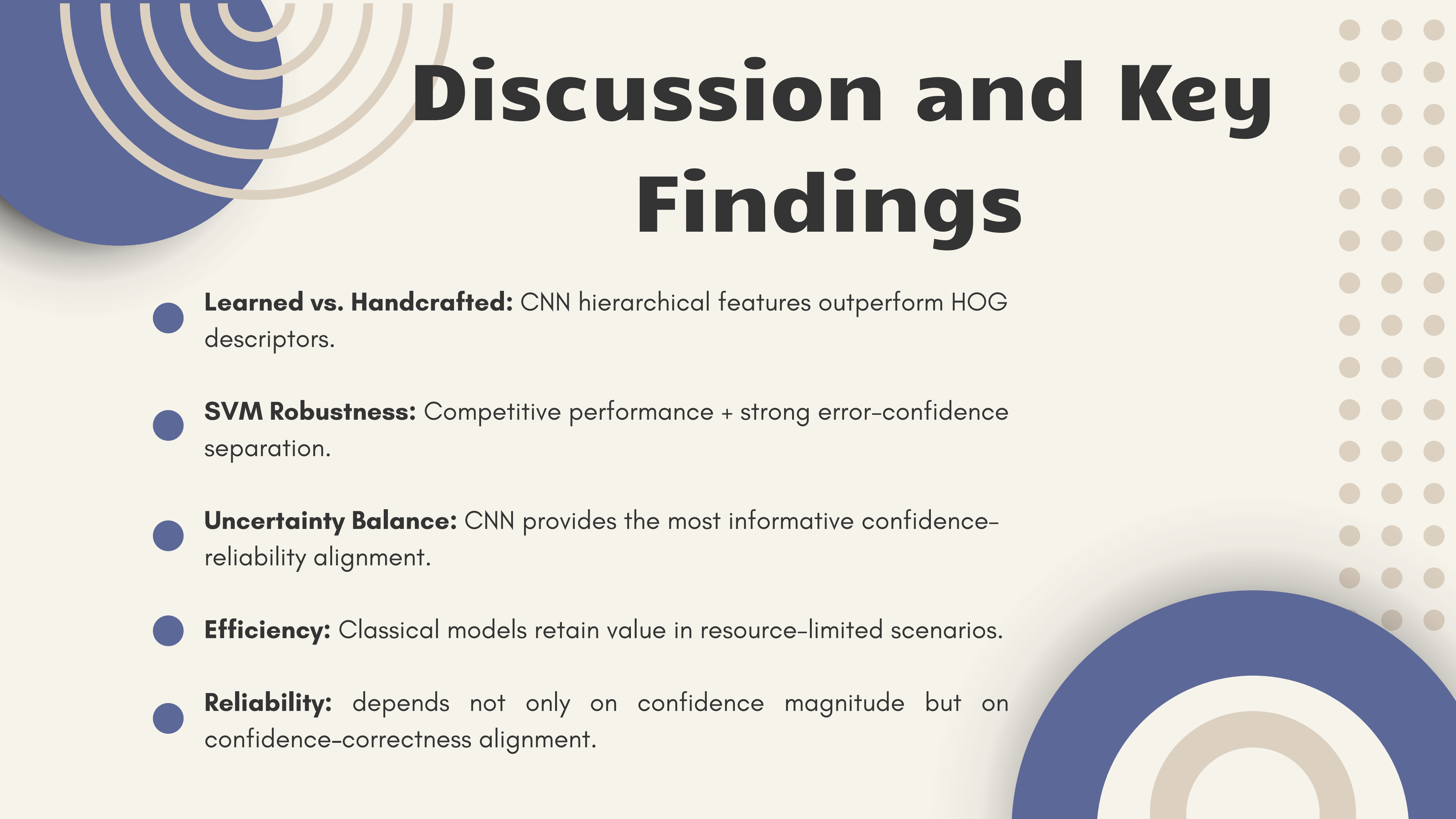
SVM : Remained competitive (QWK=0.85, F1=0.80) and achieved the highest E-AUROC (0.86), showing strong error-confidence separability.



RF: Underperformed relative to both models; however, applying PCA improved its QWK from 0.74 to 0.78.







Discussion and Key Findings

- **Learned vs. Handcrafted:** CNN hierarchical features outperform HOG descriptors.
- **SVM Robustness:** Competitive performance + strong error-confidence separation.
- **Uncertainty Balance:** CNN provides the most informative confidence-reliability alignment.
- **Efficiency:** Classical models retain value in resource-limited scenarios.
- **Reliability:** depends not only on confidence magnitude but on confidence-correctness alignment.



Conclusion

- **Summary:** Deep Learning offers superior accuracy and reliability for clinical support.
- **Baseline:** HOG + SVM is a robust, efficient alternative for limited environments.
- **Uncertainty estimation:** Confidence-aware evaluation is essential for safe medical decision-support systems.
- **Limitations:** The study was evaluated solely on the APTOS 2019 dataset.
- **Future Work:** External validation (EyePACS, Messidor) and advanced uncertainty methods.

References:

- [1] Kaggle, APTOS 2019 Blindness Detection Dataset.
 - [2] V. Gulshan et al., JAMA, 2016.
 - [3] D. Sarwinda et al., HOG-based DR classification, 2018.
 - [4] N. Dalal, B. Triggs, HOG descriptor, CVPR, 2005.
 - [5] M. Akram et al., Uncertainty-aware DR detection, 2025.
- And others.
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